FNCE30007 Mid-sem test B

Derivative Securities (University of Melbourne)

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Student Number:
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THE UNIVERSITY OF MELBOURNE
DEPARTMENT OF FINANCE

Mid-Semester Test—SEMESTER 1, 2024

SUBJECT NUMBER:	FNCE30007
SUBJECT NAME:	Derivative Securities
EXAMINATION DURATION:	1 hour
READING TIME:	5mins
TOTAL MARKS:	15
NO. OF PAGES:	9

(Do NOT write your name anywhere on this exam booklet)

AUTHORISED MATERIALS

1. This is a CLOSED BOOK examination.
2. Only Casio FX82 (any suffix) calculators are permitted. No other models of calculators will be permitted in the exam.
3. Students are NOT permitted to use any dictionary.

INSTRUCTIONS TO INVIGILATORS

1. Students should be provided ONLY with this booklet and an M1 Multiple Choice Answer Sheet.
2. Writing is NOT permitted during reading time.
3. This booklet must be COLLECTED at the end of the examination.

INSTRUCTIONS TO STUDENTS

1. Writing is NOT permitted during reading time.
2. Once writing is permitted, write your seat number and student number in the boxes at the top of this cover page.
3. Complete the M1 Multiple Choice Answer Sheet as follows:
   - Write your subject code and surname in the boxes provided on the sheet.
   - Write your Student Number under the words Student Number and then colour in the corresponding circles beneath the number. Do not put zeros at the beginning of the student ID on the multiple choice form. Begin with the first digit of your student ID.
4. Answer ALL questions. Record your answers on the M1 Multiple Choice Answer Sheet. Answers recorded in any other space will not be marked.
5. Pages titled "THIS PAGE HAS BEEN LEFT INTENTIONALLY BLANK" are provided for rough work. Examiners will NOT take into account anything written on these pages.
6. A formula list is provided on the last page. You may detach this page.
7. Do NOT detach any other pages.
8. **This paper must be handed in at the end of the examination. Do NOT remove this paper from the examination room.**

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1. Which of the following is NOT true about call and put options:
    A. An American option can be exercised at any time during its life
    B. A European option can only be exercised on the maturity date
    C. Investors must pay an upfront price (the option premium) for an option contract
    D. The price of a call option increases as the strike price increases
    E. None of the above

2. The price of a stock on July 1 is $57. A trader buys 100 call options on the stock with a strike price of $60 when the option price is $2. The options are exercised when the stock price is $65. The trader's net profit is
    A. $700
    B. $500
    C. $300
    D. $600
    E. None of the above

    (5-2)*100 = 300

3. A trader is short a portfolio worth $5 million that mirrors the performance of a stock index. The stock index is currently 1,250. Futures contracts trade on the index with one contract being on $25 times the index. To remove market risk from the portfolio the trader should
    A. Buy 160 contracts
    B. Sell 160 contracts
    C. Buy 200 contracts
    D. Sell 200 contracts
    E. None of the above

    5million/(1250*25) = buy 160 contracts

4. A company enters into a long futures contract to buy 1,000 units of a commodity for $60 per unit. The initial margin is $6,000 and the maintenance margin is $4,000. What futures price will allow $2,000 to be withdrawn from the margin account?
    A. $58
    B. $62
    C. $64
    D. $66
    E. None of the above

5. When the time to maturity increases with all else remaining the same, which of the following is true?
    A. European options always increase in value
    B. The value of European options either stays the same or increases
    C. There is no effect on European option values
    D. European options may increase or decrease in value
    E. None of the above

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6. Which of the following describes a situation where an American put option on a stock becomes more likely to be exercised early?
    A. Expected dividends increase
    B. Interest rates decrease
    C. The stock price volatility decreases
    D. All of the above
    E. None of the above

7. Interest rates are zero. A European call with a strike price of $50 and a maturity of one year is worth $6. A European put with a strike price of $50 and a maturity of one year is worth $7. The current stock price is $49. Which of the following is true?
    A. The call price is high relative to the put price
    B. The put price is high relative to the call price
    C. Both the call and put must be mispriced
    D. Both the call and put are priced correctly
    E. None of the above

8. A short forward contract that was negotiated some time ago will expire in three months and has a delivery price of $40. The current forward price for three-month forward contract is $42. The three month risk-free interest rate (with continuous compounding) is 8%. What is the value of the short forward contract (to 2 decimal places)?

    A. +$2.00
    B. −$2.00
    C. +$1.96
    D. −$1.96
    E. None of the above

9. What should an arbitrageur do when the one-year forward price of an asset is too low? Assume that the asset provides no income.
    A. The arbitrageur should borrow the price of the asset, buy one unit of the asset and enter into a short forward contract to sell the asset in one year.
    B. The arbitrageur should borrow the price of the asset, buy one unit of the asset and enter into a long forward contract to buy the asset in one year.
    C. The arbitrageur should short the asset, invest the proceeds of the short sale at the risk-free rate, enter into a short forward contract to sell the asset in one year
    D. The arbitrageur should short the asset, invest the proceeds of the short sale at the risk-free rate, enter into a long forward contract to buy the asset in one year
    E. None of the above

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10. The basis is defined as spot minus futures. A trader is hedging the purchase of an asset with a long futures position. The basis increases unexpectedly. Which of the following is true?
    A. The hedger's position improves.
    B. The hedger's position worsens.
    C. The hedger's position sometimes worsens and sometimes improves.
    D. The hedger's position stays the same.
    E. None of the above

11. On March 1 the price of a commodity is $1,000 and the December futures price is $1,015. On November 1 the price is $980 and the December futures price is $981. A producer of the commodity entered into a December futures contracts on March 1 to hedge the sale of the commodity on November 1. It closed out its position on November 1. What is the effective price (after taking account of hedging) received by the company for the commodity?
    A. $1,016
    B. $1,001
    C. $981
    D. $1,014
    E. None of the above

12. Which of the following increases basis risk?
    A. A large difference between the futures prices when the hedge is put in place and when it is closed out
    B. Dissimilarity between the underlying asset of the futures contract and the hedger's exposure
    C. A reduction in the time between the date when the futures contract is closed and its delivery month
    D. All of the above
    E. None of the above

13. A company will buy 2000 units of a certain commodity on June 30. It decides to hedge 70% of its exposure using futures contracts. The spot price is currently $105 and the August contract on the commodity is priced at $92. If the spot price and the futures price on June 30 turn out to be $108 and $103, respectively. What is the average price paid for the commodity?
    A. $97
    B. $100.3
    C. $119
    D. $115.7
    E. None of the above

108-(103-92)=97
97*0.7 + 108*0.3=100.3

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400e^(0.08-0.06)*(4/12)

14. Suppose that the risk-free rate is 8% per annum with continuous compounding and that the dividend yield on a stock index is 6% per annum. The index is standing at 400, and the futures price for a contract deliverable in four months is 405. What arbitrage opportunities, including positions to be taken in both the futures and spot markets, does this create? (to 2 decimal places)?
    A. Go long on the futures contract and short the stock underlying the index to lock in a profit of 2.32.
    B. Go short on the futures contract and long the stock underlying the index to lock in a profit of 2.32.
    C. Go long on the futures contract and long the stock underlying the index to lock in a profit of 2.32.
    D. Go short on the futures contract and short the stock underlying the index to lock in a profit of 2.32.

15. A company has a $36 million portfolio with a beta of 1.2. The futures price for a contract on an index is 9000. Futures contracts on $25 times the index can be traded. What trade is necessary to increase beta to 1.8?
    A. Long 192 contracts
    B. Short 192 contracts
    C. Long 96 contracts
    D. Short 96 contracts
    E. None of the above

(1.8-1.2)*(36m/9000*25)=

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FORMULAE SHEET

Cost of Asset/ Price Realised = $F_0 + (S_1 - F_1) = F_0 + \text{Basis}_1$

$F_0 + (S^*_1 - F_1) + (S_1 - S^*_1)$

$$N^* = \beta \frac{P}{A}$$

$$N^* = (\beta - \beta^*) \frac{P}{A}$$

$$r_c = m \cdot ln\left(1 + \frac{r_m}{m}\right)$$

$$F_0 = S_0 e^{rT}$$

$$F_0 = (S_0 + Q)e^{rT}$$

$$F_0 = S_0 e^{(r+q)T}$$

$$F_0 = (S_0 - D)e^{rT}$$

$$F_0 = S_0 e^{(r-d)T}$$

$$F_0 = S_0 e^{(r-r_f)T}$$

$$F_0 = S_0 e^{(r+q-y)T}$$

$$f = (F - K)e^{-rT}$$

$$f = (K - F)e^{-rT}$$

$c \leq S_0$

$C \leq S_0$

$p \leq Ke^{-rT}$

$P \leq K$

$c \geq max(S_0 - Ke^{-rT}, 0)$

$p \geq max(Ke^{-rT} - S_0, 0)$

$$c + Ke^{-rT} = p + S_0$$

$$c \geq \max(S_0 - D - Ke^{-rT}, 0)$$

$$p \geq \max(D + Ke^{-rT} - S_0, 0)$$

$S_0 - K < C - P < S_0 - Ke^{-rT}$

$c + D + Ke^{-rT} = p + S_0$

$S_0 - D - K < C - P < S_0 - Ke^{-rT}$

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